

UMF Trial S3M0S1 — Index 2: Context Relationships v1

Document version: v1 (populated index; standalone) **Date:** 27 May 2026 (AEST) **Working language:** Australian English (per standing rule v0.35) **Page format target:** A4 **Trial:** UMF Trial S3M0S1 **Author of trial:** Stephen L Browne **Drafter:** Claude Opus 4.7 Adaptive (Anthropic)

Status: First populated instance of Index 2 per design specification *UMF Trial S3M0S1 Indexes Design Spec v3.md* (Step 2b deliverable). Anchored at Amendment 11 (to be produced in Step 2d). Companion to Index 1 v1 (*UMF Trial S3M0S1 Index 1 Context Definition_v1.md*). Standalone navigation-layer deliverable.

Schema reference: Indexes Design Spec v3 §4 (bilateral edges) + §4.8 (joint-citation clusters).

§1 Purpose

Typed edge list of relationships between papers, frameworks, and concepts across UMF Trial S3M0S1 corpus. §3 covers bilateral edges (CR-NNNN). §4 covers joint-citation clusters (CR-JC-NNNN), the hyperedge structures deployed in the Scaffold against single UMF commitments.

Cross-references implemented per Spec v3 §7 hyperlinking provisions; external corpus links per §7.8.

§2 Counts at v1 close

Structure	Count
Bilateral edges (§3)	52
Joint-citation clusters (§4)	13
Total relational entries	65

Bilateral edges by family:

Family	Count
Foundational	11
Programme-coordinated	3
Framework-class	9
Structural	14
Citational	4
Parameter-regime	8
Modal-tier	3
Total	52

§3 Bilateral edges

§3.1 Foundational edges

CR-0001

Source: Kofinas-Saridakis-2014 **Target:** F-05/Samaddar-Singh-2025 **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** Kofinas-Saridakis 2014 *Phys. Rev. D* 90:084044 is the foundational f(T, TG) gravity construction underpinning Samaddar-Singh 2025's teleparallel-Gauss-Bonnet branch deployment. Cited via Samaddar-Singh bibliography ref [45]. **Source provenance:** Amendment 10 §3.H; F-05Paper3Samaddar2025AdjacencyReview2026_deeppass **First introduced:** Amendment 10 (27 May 2026) **Status:** Active **Cross-references:** [CD-OP-026](#) (Index 1 — informative modal status)

CR-0002

Source: F-05/Davoudiasl-2004 **Target:** F-05/Pereira-2026 **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** Davoudiasl 2004 *PRL* 93:201301 is the foundational gravitational baryogenesis mechanism that Pereira 2026 STB extends and partially addresses for stability problems (per Arbuzova et al. 2023 stability-problems flag — see CR-0030). **Source provenance:** Scaffold §5.2; F-05Paper2Pereira2026PostJan2026deeppass **First introduced:** v7 (Amendment 9) **Status:** Active

CR-0003

Source: F-05/Davoudiasl-2004 **Target:** F-05 cluster (collective) **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** Davoudiasl 2004 is the foundational reference for the entire F-05 modified-gravity-extensions sub-list. All cluster members extend the original GB mechanism through different routes (modified gravity, scalar-tensor completion, teleparallel formulation). **Source provenance:** Scaffold §5.2 **First introduced:** Pre-trial **Status:** Active

CR-0004

Source: A-01/Wheeler-1955 **Target:** A-05/Liebling-Palenzuela-2023 **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** A-05 boson stars / oscillons cited as modern realisation of A-01 Wheeler-geon programme per explicit historical-lineage attribution in Liebling-Palenzuela 2023. **Source provenance:** Scaffold §3.2 **First introduced:** Pre-trial **Status:** Active

CR-0005

Source: A-01/Wheeler-1955 **Target:** Dzhunushaliev-Folomeev-2025 **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** Dzhunushaliev-Folomeev 2025 arXiv:2512.00958 carries verbatim Wheeler attribution: "Such a configuration incarnates Wheeler's idea of 'mass without mass' and 'charge without charge'". Active 2025 instance of the Wheeler-tradition lineage. **Source provenance:** Scaffold §3.2 **First introduced:** v5 **Status:** Active

CR-0006

Source: B-04/Wheeler-1990-lfB **Target:** F-08/Verlinde-EG **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** F-08 Verlinde emergent gravity is modern technical realisation of B-04 "It from Bit" programme via Bekenstein-Hawking entropy → 't Hooft 1993 holographic principle → Susskind 1995 → Jacobson 1995 → Verlinde 2010/2017 lineage. Foundational connection per Scaffold §3.2. **Source provenance:** Scaffold §3.2, §4.4 **First introduced:** Pre-trial **Status:** Active

CR-0007

Source: B-02/Brill-1959 **Target:** UMF-Gshock **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** Brill 1959 provides the rigorous mathematical foundation for matter from gravity alone (pure vacuum GW packets carry positive ADM mass-energy with no matter fields). One of three pillars of UMF G-shock matter-creation

precedent structure. **Source provenance:** Scaffold §5.1 **First introduced:** Pre-trial **Status:** Active

CR-0008

Source: F-04/cosmic-strings **Target:** UMF-Gshock **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** F-04 cosmic strings demonstrate clean pure matter-creation from coherent topological energy via network evolution. One of three pillars of UMF G-shock matter-creation precedent structure (Copeland & Kibble 2010). **Source provenance:** Scaffold §5.1 **First introduced:** Pre-trial **Status:** Active

CR-0009

Source: B-03/ADD-1998 **Target:** UMF-A4 **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** ADD 1998 large extra dimensions provides the framework for G-variation under variant geometry ($G_4 = G(4+n)/V_n$). Foundational for UMF v0.7 §3.1 assumption 4 G-variation admission. **Source provenance:** Scaffold §4.2 **First introduced:** Pre-trial **Status:** Active

CR-0010

Source: B-01/Misner-Wheeler-1957 **Target:** UMF-A4 **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** B-01 Misner-Wheeler 1957 (Rainich 1925 already-unified construction; EM field tensor $F_{\mu\nu}$ derived from Ricci tensor $R_{\mu\nu}$) is the closest mainstream framework to UMF v0.7 §3.1 assumption 4 ϵ_0/μ_0 -coupled-to-geometry admission identified in trial. **Source provenance:** Scaffold §4.3 **First introduced:** Pre-trial; Stephen Q3 deployment 25 May 2026 **Status:** Active **Cross-references:** [CD-OP-Adj-003](#)

CR-0011

Source: F-07/Sorkin-1990 **Target:** Λ -DE-discovery-1998 **Edge type:** framework-foundational **Edge type family:** Foundational **Symmetry:** Asymmetric **Annotation:** Sorkin 1990 causal-set theory prediction $\Lambda \sim 1/\sqrt{V}$ universe predates and matches order-of-magnitude the 1998 dark-energy discovery. Strongest successful pre-discovery prediction in audit corpus (not 5σ but notable). **Source provenance:** Scaffold §9.3, §10.2 **First introduced:** Pre-trial **Status:** Active

§3.2 Programme-coordinated edges

CR-0012

Source: F-05/Malakar-2025 (Paper 1) **Target:** F-05/Malakar-Mazumdar-Bhuyan-2026 (Paper 2) **Edge type:** programme-coordinated **Edge type family:** Programme-coordinated **Symmetry:** Symmetric **Annotation:** Two papers in coordinated Malakar-group research programme on modified-gravity baryogenesis. Paper 1 Nojiri-Odintsov $f(R)$; Paper 2 $f(R, G_{\mu\nu} T^{\mu\nu})$ EMSG-adjacent. Together with Paper 4 (Malakar-Mazumdar-Gohain-Bhuyan 2026 $f(R, L_m, T)$) constitute three-paper group span June 2025 – May 2026. **Source provenance:** Amendment 10 §3.E; Scaffold §5.2 [v8 addition] **First introduced:** Amendment 10 (27 May 2026) **Status:** Active

CR-0013

Source: F-05/Malakar-2025 (Paper 1) **Target:** F-05/Malakar-Mazumdar-Gohain-Bhuyan-2026 (Paper 4 of original review list) **Edge type:** programme-coordinated **Edge type family:** Programme-coordinated **Symmetry:** Symmetric **Annotation:** Same Malakar-group research programme on modified-gravity baryogenesis. Paper 4 $f(R, L_m, T)$ integrated v7 (Amendment 9 cycle); Paper 1 integrated v8 (Amendment 10 AdjacencyReview2026 cycle). Group span confirms coordinated programme. **Source provenance:** Amendment 10 §3.E **First introduced:** Amendment 10 **Status:** Active

CR-0014

Source: F-05/Cruz-Pereira-Lobo-2025 **Target:** Marciu-2026 **Edge type:** programme-coordinated **Edge type family:** Programme-coordinated **Symmetry:** Symmetric **Annotation:** Same Lobo/Lisbon group adjacency. Cruz-Pereira-Lobo 2025/2026 $f(T, L_m)$ and Marciu 2026 EPJC 86:543 EMSG scalar-tensor representation share institutional and methodological adjacency. Pereira 2026 (Paper 2 of original review list) also Lobo group adjacent. **Source provenance:** Scaffold §5.2 [v7

addition] **First introduced:** v7 (Amendment 9 cycle) **Status:** Active

§3.3 Framework-class edges

CR-0015

Source: F-05/Davoudiasl-2004 **Target:** F-05/Lambiase-Scarpetta-2006 **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Symmetric **Annotation:** Both members of F-05 curvature-based modified-gravity-extensions branch. Lambiase-Scarpetta 2006 *PRD* 74:087504 $f(R)$ extension. **Source provenance:** Scaffold §5.2 **First introduced:** Pre-trial **Status:** Active

CR-0016

Source: F-05/Pereira-2026 **Target:** F-05/Whittingham-2026 **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Symmetric **Annotation:** Both members of F-05 curvature-based branch. Pereira STB scalar-tensor completion; Whittingham Einstein-frame scalaron approach for Starobinsky and Odintsov-Oikonomou 2025 power-law $f(R)$. Different specific approaches within same model class. **Source provenance:** Scaffold §5.2 [v7 addition] **First introduced:** v7 (Amendment 9 cycle) **Status:** Active

CR-0017

Source: F-05/Pereira-2026 **Target:** F-05/Malakar-Mazumdar-Gohain-Bhuyan-2026 (Paper 4) **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Symmetric **Annotation:** Both F-05 curvature-based branch members (different specific f -functions). **Source provenance:** Scaffold §5.2 [v7 addition] **First introduced:** v7 **Status:** Active

CR-0018

Source: F-05/Cruz-Pereira-Lobo-2025 **Target:** F-05/Samaddar-Singh-2025 **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Symmetric **Annotation:** Both members of F-05 teleparallel branch. Cruz-Pereira-Lobo $f(T, L_m)$; Samaddar-Singh $f(T, TG)$. Together establish the teleparallel-branch as a populated sub-cluster of F-05 (two entries at v8 close). **Source provenance:** Amendment 10 §3.E; Scaffold §5.2 [v8 addition] **First introduced:** v8 (Amendment 10) **Status:** Active

CR-0019

Source: F-05/Malakar-Mazumdar-Bhuyan-2026 (Paper 2) **Target:** Marciu-2026 **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Symmetric **Annotation:** Both EMSG-adjacent. Malakar Paper 2 uses $f(R, G_{\mu\nu} T^{\mu\nu})$ where $G_{\mu\nu}$ is Einstein tensor. Marciu 2026 *EPJC* 86:543 energy-momentum squared gravity in scalar-tensor representation. **Source provenance:** Amendment 10 §3.E **First introduced:** v8 (Amendment 10) **Status:** Active

CR-0020

Source: F-02/Liu-Qin-Bian-2026 **Target:** F-05 cluster (collective) **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Asymmetric **Annotation:** NOT shared — Liu-Qin-Bian 2026 is F-02 (first-order phase transition) alternative class, distinct from F-05 gravitational baryogenesis. Edge recorded as **negative** framework-class relationship (corpus-discipline distinction). **Source provenance:** Scaffold §5.2 [v7 addition] **First introduced:** v7 **Status:** Active

CR-0021

Source: Michiwaki-Kobayashi-2026 **Target:** F-05/Pereira-2026 **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Asymmetric **Annotation:** Michiwaki-Kobayashi 2026 Generalised Disformal Horndeski scalar-tensor theories with third-order derivatives provides structural-mathematical framework underpinning Pereira 2026 STB. Adjacent structural-mathematical contribution to F-05 cluster. **Source provenance:** Scaffold §5.2 [v7 addition] **First introduced:** v7 **Status:** Active

CR-0022

Source: A-01/Wheeler-1955 **Target:** A-02/pilot-wave-class **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Symmetric **Annotation:** Both Theme C entries within GWM particle-representation three-way structure (with A-09 mathematical). A-01 historical / A-02 pilot-wave subclass relationship per Scaffold §8. **Source provenance:** Scaffold §8.1, §8.3 **First introduced:** Pre-trial **Status:** Active

CR-0023

Source: F-06/LQG **Target:** F-07/CST **Edge type:** framework-class-shared **Edge type family:** Framework-class **Symmetry:** Symmetric **Annotation:** Both quantum-gravity matter-coupling-open methodological precedents. Joint citation deployed at Scaffold §9.1. **Source provenance:** Scaffold §9.1 **First introduced:** Pre-trial **Status:** Active

§3.4 Structural edges

CR-0024

Source: F-05/Pereira-2026 **Target:** F-05/Davoudiasl-2004 **Edge type:** extends **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Pereira 2026 STB scalar-tensor completion extends Davoudiasl 2004 via two formal propositions providing explicit on-shell/background dictionary (Legendre relations $\xi_i = V\{\Phi_i\}$); *three selection rules (branch existence, local invertibility, spectator validity)*. *Structural organising result for F-05 cluster*. **Source provenance:** Scaffold §5.2 [v7 addition]; F-05Paper2Pereira2026PostJan2026_deeppass **First introduced:** v7 **Status:** Active

CR-0025

Source: F-05/Whittingham-2026 **Target:** F-05/Starobinsky-f(R) **Edge type:** refines **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Whittingham 2026 finds that with $M^* = MPI$, both Starobinsky and Odintsov-Oikonomou 2025 power-law $f(R)$ models yield $\eta \approx (1.05-1.53) \times 10^{-11}$ — approximately 5-8x below observed value, requiring $M^*/MPI \approx 0.4$ for agreement. Honest-disclosure observation regarding parameter requirements. **Source provenance:** Scaffold §5.2 [v7 addition] **First introduced:** v7 **Status:** Active

CR-0026

Source: F-05/Whittingham-2026 **Target:** F-05/Odintsov-Oikonomou-2025-power-law-f(R) **Edge type:** refines **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Same as CR-0025 for Odintsov-Oikonomou 2025 power-law $f(R)$ model. η -undershoot finding applies. **Source provenance:** Scaffold §5.2 [v7 addition] **First introduced:** v7 **Status:** Active

CR-0027

Source: F-02/Liu-Qin-Bian-2026 **Target:** SM-Higgs-sector **Edge type:** extends **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Liu-Qin-Bian 2026 EPJC extends SM with dimension-6 CPV operator $(\Phi^\dagger\Phi)B_{\mu\nu}\tilde{B}^{\mu\nu}$. *Mechanism: helical magnetic field generation in two-bubble collisions during first-order EWPT; baryon asymmetry via Chern-Simons number variation*. EDM constraint (Roussy 2023, $d_e/e < 4.1 \times 10^{-30}$ cm) limits $\Lambda CPV \gtrsim 338$ TeV; CPV-operator-alone gives $\eta B \approx 1.8 \times 10^{-14}$ (7 OoM below observed). **Source provenance:** Scaffold §5.2 [v7 addition]; F-02Paper8Liu2026PostJan2026_deeppass **First introduced:** v7 **Status:** Active

CR-0028

Source: Mendoza-Hernandez-2014 **Target:** Λ CDM-Bullet-Cluster-interpretation **Edge type:** contests **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Mendoza-Hernandez et al. 2014 (arXiv:1401.7063) verbatim: "the inferred infall velocity of the bullet cluster is inconsistent with the standard cosmological scenario". Λ CDM collision-velocity problem (~ 4000 km/s natural in MOND/MOG, problematic in Λ CDM). **Source provenance:** Scaffold §10.2 **First introduced:** v3 (Bullet Cluster correction) **Status:** Active

CR-0029

Source: Brownstein-Moffat-2007 **Target:** Collisionless-DM-interpretation **Edge type:** contests **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Brownstein & Moffat 2007 MNRAS 382:29 verbatim: "the 'normal' matter in the Bullet

Cluster is enough to account for the observed gravitational lensing effect". MOG/STVG alternative interpretation of Bullet Cluster phenomenon. **Source provenance:** Scaffold §10.2 **First introduced:** v3 **Status:** Active

CR-0030

Source: Arbuzova-2023 **Target:** F-05/Davoudiasl-2004 **Edge type:** contests **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Arbuzova et al. 2023 *Symmetry* 15(2):404 "Gravitational baryogenesis: problems and possible resolution" flags known stability problems with original Davoudiasl 2004 GB. Surfaced via Pereira 2026 bibliography; Pereira STB claims to partially address. **Source provenance:** Scaffold §5.2 [v7 honest disclosure]; F-05Paper2Pereira2026PostJan2026deeppass **First introduced:** v7 **Status:** Active

CR-0031

Source: Rusakov-2026 **Target:** LRD-anomaly-frameworks **Edge type:** contests **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Rusakov et al. 2026 *Nature* 649:574-579 standard-physics resolution of Little Red Dots identity (electron-scattering-cocoon mechanism; mass revision into Reines-Volonteri 2015 scaling). Resolves LRDs within standard astrophysics; reduces empirical motivation for non-standard-physics frameworks specifically for LRD category. JWST category (a) impossibly-early + (c) massive-galaxy number-density remain open. **Source provenance:** Scaffold §10.4.1(b); v4 honest-disclosure correction **First introduced:** v4 (post-Rusakov correction) **Status:** Active

CR-0032

Source: A-05/boson-stars **Target:** A-01/Wheeler-1955 **Edge type:** extends **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Liebling-Palenzuela 2023 explicit historical-lineage attribution: A-05 boson stars / oscillons as modern realisation of A-01 Wheeler-geon programme. Modern resolution of A-01 classical-stability concerns. **Source provenance:** Scaffold §3.2 **First introduced:** Pre-trial **Status:** Active

CR-0033

Source: F-05/Pereira-2026 Proposition framework **Target:** F-05 cluster (collective) **Edge type:** refines **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Pereira 2026 two propositions establish on-shell/background dictionary between curvature-based geometric GB operators and scalar-tensor counterparts. Three selection rules constrain admissible operator space. Structural organising result for F-05 cluster (not merely an additional extension reference per F-05Paper2Pereira2026PostJan2026deeppass). **Source provenance:** Scaffold §5.2 [v7 addition]; §11.3 Item 5 **First introduced:** v7 **Status:** Active **Cross-references:** [CD-OP-Item-005](#)

CR-0034

Source: F-05/Samaddar-Singh-2025 **Target:** F-05/Whittingham-2026 **Edge type:** parallel **Edge type family:** Structural **Symmetry:** Symmetric **Annotation:** Both arrive at modified-gravity baryogenesis branches of the F-05 cluster in similar timeframe but in distinct branches (teleparallel vs curvature-based). Independent — Samaddar-Singh does not cite Whittingham; Whittingham does not cite Samaddar-Singh. Parallel relationship. **Source provenance:** Amendment 10 §3.E; Scaffold §5.2 [v8 addition] **First introduced:** v8 (Amendment 10) **Status:** Active

CR-0035

Source: F-05/Samaddar-Singh-2025 **Target:** F-05/Pereira-2026 **Edge type:** parallel **Edge type family:** Structural **Symmetry:** Symmetric **Annotation:** Both in F-05 cluster but distinct branches and independently developed. Samaddar-Singh reference-list-isolated from contemporaneous parallel cluster work (does not cite Pereira 2026 *Phys. Lett. B* 875:140312 nor Cruz-Pereira-Lobo 2025/2026 nor Goodarzi 2026 despite predating December 2025 submission). **Source provenance:** Amendment 10 §3.G; F-05Paper3Samaddar2025AdjacencyReview2026deeppass §7.9 **First introduced:** v8 (Amendment 10) **Status:** Active

CR-0036

Source: Bullet-Cluster-phenomenon **Target:** Bullet-Cluster-interpretations **Edge type:** orthogonal **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Phenomenon-vs-interpretation distinction: 8σ measurement (Clowe et al. 2006 *ApJ*)

648:L109) for general mass-gravity-offset phenomenon is observed regardless of interpretive framework. Multiple competing interpretations (collisionless DM, MOG/STVG, TeVeS, Refracted Gravity, Randers-Finslerian, Mashhoon) have no 5σ confirmation as uniquely correct. Honest-disclosure correction #1. **Source provenance:** Scaffold §10.2; v3 honest-disclosure correction **First introduced:** v3 **Status:** Active **Cross-references:** [CD-OP-012](#) (honest-disclosure correction #1)

CR-0037

Source: F-05/Goodarzi-2026 **Target:** F-05/Davoudiasl-2004 **Edge type:** extends **Edge type family:** Structural **Symmetry:** Asymmetric **Annotation:** Goodarzi 2026 *JHEP* 02:029 F(R) gravity's rainbow baryogenesis extends Davoudiasl mechanism. Peer-reviewed F-05 cluster member integrated v7. **Source provenance:** Scaffold §5.2 [v7 addition] **First introduced:** v7 **Status:** Active

§3.5 Citational edges

CR-0038

Source: F-05/Samaddar-Singh-2025 **Target:** Kofinas-Saridakis-2014 **Edge type:** cites **Edge type family:** Citational **Symmetry:** Asymmetric **Annotation:** Samaddar-Singh ref [45]. Foundational citation for $f(T, TG)$ gravity construction; relationship also captured as foundational edge CR-0001. **Source provenance:** F-05Paper3Samaddar2025AdjacencyReview2026_deeppass **First introduced:** v8 (Amendment 10) **Status:** Active

CR-0039

Source: F-05/Samaddar-Singh-2025 **Target:** Azhar-Jawad-Rani-2020 **Edge type:** cites **Edge type family:** Citational **Symmetry:** Asymmetric **Annotation:** Samaddar-Singh cites Azhar-Jawad-Rani 2020 *Phys. Dark Univ.* 30:100724 as prior $f(T, TG)$ baryogenesis result acknowledged but NOT quantitatively compared (honest-disclosure observation §7.10). Azhar-Jawad-Rani 2020/2021 held aside per Amendment 10 §3.I. **Source provenance:** Amendment 10 §3.I; F-05Paper3Samaddar2025AdjacencyReview2026_deeppass §7.10 **First introduced:** v8 (Amendment 10) **Status:** Active; held-aside-context note: target nodes Azhar-Jawad-Rani 2020/2021 currently Held-aside per Amendment 10 §3.I **Held-aside reason:** Target papers held aside pending Stephen direction (Handoff v6 §12 item (b))

CR-0040

Source: F-05/Samaddar-Singh-2025 **Target:** F-05 contemporaneous-cluster-work **Edge type:** is-cited-by **Edge type family:** Citational **Symmetry:** Asymmetric **Annotation:** Negative citational edge (reference-list isolation): Samaddar-Singh 2025 does NOT cite contemporaneous 2025 parallel cluster work — arXiv:2509.03053 (Malakar Sep 2025; Paper 4); arXiv:2509.17218 (Cruz-Pereira-Lobo Sep 2025); arXiv:2508.16955 (Goodarzi Aug 2025) — all predating 3 Dec 2025 submission. Honest-disclosure observation §7.9. **Source provenance:** Amendment 10 §3.E; F-05Paper3Samaddar2025AdjacencyReview2026deeppass §7.9 **First introduced:** v8 (Amendment 10) **Status:** Active

CR-0041

Source: F-05/Pereira-2026 **Target:** Arbuzova-2023 **Edge type:** cites **Edge type family:** Citational **Symmetry:** Asymmetric **Annotation:** Pereira 2026 bibliography surfaces Arbuzova et al. 2023 *Symmetry* 15(2):404 stability-problems flag. STB framing claims to partially address. **Source provenance:** F-05Paper2Pereira2026PostJan2026deeppass **First introduced:** v7 **Status:** Active

§3.6 Parameter-regime edges

CR-0042

Source: F-05/Pereira-2026 (M^*) **Target:** F-05/Whittingham-2026 (M^*) **Edge type:** parameter-regime-different **Edge type family:** Parameter-regime **Symmetry:** Symmetric **Annotation:** Pereira $M^* = MPI$ with $\varepsilon \approx O(10^{-6})$ tuning. Whittingham resolution requires $M^*/MPI \approx 0.4$ (factor-2.4 reduction). Different M^* regimes within curvature-based branch. **Source provenance:** Scaffold §5.2 [v7 addition]; §11.3 Item 5 **First introduced:** v7 **Status:** Active **Cross-references:** [MU-002](#)

CR-0043

Source: F-05/Whittingham-2026 (M*) **Target:** F-05/Samaddar-Singh-2025 (M*) **Edge type:** parameter-regime-different **Edge type family:** Parameter-regime **Symmetry:** Symmetric **Annotation:** Whittingham $M/M_{PI} \approx 0.4$. Samaddar-Singh $M = 2 \times 10^{12}$ GeV $\approx 10^{-7} M_{PI}$. Difference of ~ 6.4 orders of magnitude. **Source provenance:** Amendment 10 §3.E; Scaffold §11.3 Item 5 v8 substantive extension **First introduced:** v8 (Amendment 10) **Status:** Active **Cross-references:** [MU-002](#)

CR-0044

Source: F-05/Pereira-2026 (M*) **Target:** F-05/Samaddar-Singh-2025 (M*) **Edge type:** parameter-regime-different **Edge type family:** Parameter-regime **Symmetry:** Symmetric **Annotation:** Pereira $M^* = MPI \approx 1.22 \times 10^{19}$ GeV. Samaddar-Singh $M^* = 2 \times 10^{12}$ GeV. Cross-cluster spread ~ 7 OoM. F-05 cluster M^* spread: MPI per Pereira; ≈ 0.4 MPI per Whittingham; $\approx 10^{-7}$ MPI per Samaddar-Singh. **Source provenance:** Amendment 10 §3.E; Scaffold §11.3 Item 5 **First introduced:** v8 (Amendment 10) **Status:** Active **Cross-references:** [MU-002](#), [CD-OP-Item-005](#)

CR-0045

Source: F-05/Davoudiasl-2004 ($\eta B/s$) **Target:** F-05/Pereira-2026 ($\eta B/s$) **Edge type:** parameter-regime-different **Edge type family:** Parameter-regime **Symmetry:** Symmetric **Annotation:** Davoudiasl 2004 corpus verbatim $9.2^{+0.6}_{-0.4} \times 10^{11}$. Pereira 2026 adopts $(8.8 \pm 0.6) \times 10^{11}$. Difference reflects ~ 22 years of CMB-precision improvement; honest-disclosure correction #3 captures observational-target evolution across 2026 cluster. **Source provenance:** Scaffold §5.2 [v7 honest disclosure] **First introduced:** v7 **Status:** Active **Cross-references:** [MU-001](#), [CD-OP-012](#)

CR-0046

Source: F-05/Samaddar-Singh-2025 ($\eta B/s$) **Target:** F-05 cluster ($\eta B/s$) **Edge type:** parameter-regime-different **Edge type family:** Parameter-regime **Symmetry:** Asymmetric **Annotation:** Samaddar-Singh $\eta_{B/s} = 9.42 \times 10^{11}$ (via Planck 2016 + Kolb-Turner 1990 textbook). Slightly higher than Pereira 8.8, Whittingham 8.65 CMB / 8.2-9.4 BBN range, Liu-Qin-Bian 8.8. Within Davoudiasl 2004 corpus range 9.2 ± 0.6 . Highest observational target in 2026 cluster. **Source provenance:** Amendment 10 §3.E; Research Plan v0.6 §8.1 **First introduced:** v8 (Amendment 10) **Status:** Active **Cross-references:** [MU-001](#)

CR-0047

Source: F-05/Davoudiasl-2004 (TD) **Target:** F-05/Samaddar-Singh-2025 (TD) **Edge type:** parameter-regime-shared **Edge type family:** Parameter-regime **Symmetry:** Symmetric **Annotation:** Both use GUT-scale $TD \sim 10^{16}$ GeV. Samaddar-Singh $TD = 2 \times 10^{18}$ GeV. Conventional in F-05 cluster. **Source provenance:** Amendment 10 §3.E; F-05Paper3Samaddar2025AdjacencyReview2026deeppass §7.6 **First introduced:** v8 (Amendment 10) **Status:** Active **Cross-references:** [CD-TH-012](#)

CR-0048

Source: F-05/Pereira-2026 (TD) **Target:** F-05 cluster (TD) **Edge type:** parameter-regime-different **Edge type family:** Parameter-regime **Symmetry:** Asymmetric **Annotation:** Pereira $TD \approx 8.5 \times 10^{13}$ GeV (below GUT scale). Differs from Samaddar-Singh and Davoudiasl conventional GUT-scale $TD \sim 10^{16}$ GeV. **Source provenance:** Scaffold §11.3 Item 5 **First introduced:** v7 **Status:** Active **Cross-references:** [CD-TH-012](#)

CR-0049

Source: F-05/Samaddar-Singh-2025 (parameter coefficients) **Target:** F-05 cluster (parameter coefficients) **Edge type:** parameter-regime-different **Edge type family:** Parameter-regime **Symmetry:** Asymmetric **Annotation:** Samaddar-Singh parameter coefficients span 10^{20} to 10^{110} orders of magnitude across models/formalisms — ~ 90 OoM spread. Paper §4 self-characterisation "no extreme fine-tuning" contradicted by displayed parameter values (honest-disclosure observation §7.3). Distinct from Pereira $\varepsilon \approx O(10^{-6})$ tuning at $M^* = MPI$. **Source provenance:** Amendment 10 §3.E; F-05Paper3Samaddar2025AdjacencyReview2026_deeppass §7.3 **First introduced:** v8 (Amendment 10) **Status:** Active **Cross-references:** [CD-TH-015](#)

§3.7 Modal-tier edges

CR-0050

Source: F-05/Whittingham-2026 (preprint-with-strong-internal-quality) **Target:** F-05/Samaddar-Singh-2025 (preprint-with-quality-concerns) **Edge type:** modal-tier-comparison **Edge type family:** Modal-tier **Symmetry:** Symmetric **Annotation:** Both preprint-only; distinct quality profiles. Whittingham: 36 pages; astro-ph.CO; explicit $\varepsilon \rightarrow 0$ Starobinsky-limit self-consistency check; "presumably submitted to JCAP". Samaddar-Singh: 14 pages; physics.gen-ph (catch-all); six manuscript-internal quality concerns; ~5 months without journal publication record. **Source provenance:** Amendment 10 §3.G; Scaffold §5.2 v8 honest disclosure **First introduced:** v8 (Amendment 10) **Status:** Active **Cross-references:** [CD-OP-019](#), [CD-OP-020](#)

CR-0051

Source: F-05/Samaddar-Singh-2025 **Target:** F-05 cluster peer-reviewed (collective) **Edge type:** modal-tier-comparison **Edge type family:** Modal-tier **Symmetry:** Asymmetric **Annotation:** Samaddar-Singh preprint-with-quality-concerns vs 9 peer-reviewed F-05 cluster members. Cross-cluster preprint-status heterogeneity honest-disclosure (honest-disclosure correction #5). Cluster has predominantly peer-reviewed source content; Samaddar-Singh sits as lower-quality outlier. **Source provenance:** Amendment 10 §3.G; Scaffold §5.2 v8 honest disclosure **First introduced:** v8 (Amendment 10) **Status:** Active **Cross-references:** [CD-OP-012](#) (honest-disclosure correction #5), [CD-OP-017](#), [CD-OP-020](#)

CR-0052

Source: F-05/Malakar-2025 (Paper 1) **Target:** F-05/Malakar-Mazumdar-Bhuyan-2026 (Paper 2) **Edge type:** modal-tier-comparison **Edge type family:** Modal-tier **Symmetry:** Symmetric **Annotation:** Both abstract-verified-only at v8 close. Journal-publication status of these two extended-verification declined this cycle per Stephen direction option 4 27 May AEST. Available for future-cycle action if authorised (Handoff v6 §12 item (a)). **Source provenance:** Amendment 10 §3.G; Scaffold §5.2 v8 honest disclosure **First introduced:** v8 (Amendment 10) **Status:** Active; available for verification extension pending Stephen direction **Cross-references:** [CD-OP-008](#)

§4 Joint citation clusters (CR-JC-NNNN)

CR-JC-0001

Cluster name: Wheeler-tradition lineage bracket **Member nodes:** A-01/Wheeler-1955, B-01/Misner-Wheeler-1957, B-02/Brill-1959, B-04/Wheeler-1990-lfB **UMF commitment supported:** Wheeler-tradition lineage adoption (Scaffold §3.1) — particles as gravitating bound geometric structures (A-01); all classical physics as curved empty space (B-01); pure gravity-only matter (B-02); information allowance compatible with It-from-Bit programme (B-04 with declined information-primary ontology) **Deployment context:** Scaffold §3 substantive section first section establishes Wheeler-tradition lineage that UMF inherits; described as "the strongest mainstream-physics historical-lineage UMF can claim" **Annotation:** Four-period Wheeler programme structure: 1st (1955) particles; 2nd (1957) geometry; 2nd application (1959) pure-gravity-only matter; 3rd (1989/1990) information. **Modal status:** committed (the lineage adoption) **Stephen adjudication ref:** N/A (pre-trial) **Source provenance:** Scaffold §3.1 **First introduced:** Pre-trial **Status:** Active

CR-JC-0002

Cluster name: Geometric-foundations programme bracket **Member nodes:** A-08/Penrose-twistor, A-09/Hestenes-spacetime-algebra, B-01/Misner-Wheeler-geometrodynamics **UMF commitment supported:** Foundational-geometric-programme legitimacy as methodological precedent for UMF's own foundational commitments **Deployment context:** Scaffold §3.3 geometric-foundations programme joint-citation **Annotation:** Penrose 2020 Nobel Prize for black-hole-formation prediction establishes ongoing mainstream-physics legitimacy of foundational geometric-programme work. **Modal status:** committed **Source provenance:** Scaffold §3.3 **First introduced:** Pre-trial **Status:** Active

CR-JC-0003

Cluster name: G-shock three-pillar precedent **Member nodes:** B-02/Brill-1959, F-04/cosmic-strings, B-03/ADD-1998 **UMF commitment supported:** UMF G-shock matter-creation mechanism **Deployment context:** Scaffold §5.1 — three independent mainstream-precedent pillars: gravity-only matter foundation (B-02); clean pure-creation mechanism (F-04); G-variation under variant geometry (B-03) **Annotation:** Three independent lines of mainstream-physics precedent jointly support UMF G-shock matter-creation. Verbatim deployment in Scaffold §5.1 cites Brill 1959, Copeland & Kibble 2010, Arkani-Hamed-Dimopoulos-Dvali 1998. **Modal status:** committed **Source provenance:** Scaffold §5.1 **First introduced:** Pre-trial **Status:** Active **Cross-references:** [CR-0007](#), [CR-0008](#), [CR-0009](#)

CR-JC-0004

Cluster name: Variant-fundamental-constants joint defence **Member nodes:** B-01/Misner-Wheeler-Rainich, B-03/ADD-1998 **UMF commitment supported:** UMF v0.7 §3.1 assumption 4 (variant fundamental constants under variant geometry: G via ADD *V_n-dependent mechanism*; ϵ_0/μ_0 via *Rainich already-unified construction*; *strong-nuclear-force coupling* as correlated reduction under same ADD *V_n-dependent mechanism*) **Deployment context:** Scaffold §4.3 **Annotation:** Combined B-01 + B-03 gives multi-citation backing for variant G + variant ϵ_0/μ_0 via different but compatible geometric mechanisms. Stephen Q3 25 May 2026 deployment direction with strong-nuclear-force context requirement. **Modal status:** committed **Stephen adjudication ref:** CD-OP-Adj-003 **Source provenance:** Scaffold §4.3 **First introduced:** Pre-trial; Stephen Q3 deployment 25 May 2026 **Status:** Active **Cross-references:** [CR-0009](#), [CR-0010](#)

CR-JC-0005

Cluster name: Information allowance **Member nodes:** B-04/Wheeler-1990-lfB, F-08/Verlinde-EG **UMF commitment supported:** UMF information-allowance position — UMF imports information-coupling aspect of It-from-Bit programme while declining information-primary ontology and participatory-universe metaphysics **Deployment context:** Scaffold §4.4 **Annotation:** B-04 historical foundation; F-08 modern technical realisation. Joint-citation deployment verbatim block in Scaffold §4.4. **Modal status:** committed (information allowance); disclaimed (information-primary ontology) **Source provenance:** Scaffold §4.4 **First introduced:** Pre-trial **Status:** Active **Cross-references:** [CR-0006](#)

CR-JC-0006

Cluster name: Five-framework gravity-mass offset **Member nodes:** F-10/Refracted-Gravity, Bullet-Cluster-phenomenon, Mashhoon-non-local-gravity, B-01/Wheeler-charge-without-charge, F-08/Verlinde-EG **UMF commitment supported:** UMF gravity-mass spatial offset mechanism — UMF gravitoelectric DM with phase-conjugate-wavefront-mediated gravity-mass offset joins competing-interpretations landscape as one additional candidate **Deployment context:** Scaffold §7.3 five-framework joint citation **Annotation:** F-10 Refracted Gravity is highest-fit framework; Bullet Cluster provides 8σ phenomenon (interpretation contested, see CR-0036); Mashhoon non-local gravity, Wheeler "charge without charge", Verlinde EG provide additional precedent. Stephen Issue 2 adjudication established the five-framework joint citation structure. **Modal status:** committed (joint citation deployed); informative (UMF specific mechanism — Item 6 open) **Stephen adjudication ref:** CD-OP-Adj-010 **Source provenance:** Scaffold §7.3 **First introduced:** v3 (Issue 2 + scope-expansion) **Status:** Active **Cross-references:** [CD-OP-Item-006](#)

CR-JC-0007

Cluster name: Compound phase-conjugate-wavefront analogue **Member nodes:** A-06/Couder-hydrodynamic, A-10/fibre-optic-horizon, Phase-conjugation-literature **UMF commitment supported:** UMF phase-conjugate wavefront commitment — reflective aspect (A-06) + transmutative aspect (phase-conjugation literature) + empirical engagement (A-10) **Deployment context:** Scaffold §7.2 **Annotation:** Stephen Issue 1 terminology revision established phase-conjugate-wavefront framing and compound A-06 + A-10 + phase-conjugation analogue joint-citation structure. A-10 fibre-optic horizon empirical engagement (Belgiorno et al. 2010 + Royal Society 2019 ~15 years). **Modal status:** committed **Stephen adjudication ref:** CD-OP-Adj-009 **Source provenance:** Scaffold §7.2 **First introduced:** v3 (Issue 1) **Status:** Active **Cross-references:** [CD-TH-018](#)

CR-JC-0008

Cluster name: Classical $\hbar$ -free gravity-only matter **Member nodes:** B-02/Brill-1959, B-01/Misner-Wheeler-1957, F-04/cosmic-strings, A-01/Wheeler-geons **UMF commitment supported:** GWM $\hbar$ -exclusion mainstream-precedent in gravity-only-matter

regime — supported by four classical-framework precedents **Deployment context:** Scaffold §5.4 Face 2 **Annotation:** Face 2 of GWM $\hbar$ -exclusion: gravity-only-matter regime substantially supported by four classical $\hbar$ -free precedents. Wheeler 1955 verbatim "completely classical, divergence-free, self-consistent picture". Distinct from Face 1 (elementary-particle regime) where E-01 closed-null finding establishes $\hbar$ -free mainstream support is absent and GWM is genuinely novel. **Modal status:** committed (precedent backing for gravity-only-matter regime); informative (elementary-particle regime requires honest disclosure of GWM novelty) **Source provenance:** Scaffold §5.4 **First introduced:** Pre-trial **Status:** Active

CR-JC-0009

Cluster name: Pilot-wave class **Member nodes:** A-02/pilot-wave-mainstream, A-06/Couder-hydrodynamic **UMF commitment supported:** GWM particle representation pilot-wave subclass **Deployment context:** Scaffold §8.3 **Annotation:** A-02 (Bohm; Dürr-Goldstein-Zanghi; Drezet-Amblard 2025; Struyve 2016) + A-06 Couder hydrodynamic walking-droplet analogue (20+ years experimental). **Modal status:** committed **Source provenance:** Scaffold §8.3 **First introduced:** Pre-trial **Status:** Active

CR-JC-0010

Cluster name: LQG + CST methodological-precedent **Member nodes:** F-06/LQG, F-07/CST **UMF commitment supported:** Quantum-gravity matter-coupling-open methodological precedent for UMF's own foundational programme **Deployment context:** Scaffold §9.1 **Annotation:** F-06 LQG (Ashtekar-Bianchi 2021); F-07 CST (Surya 2019). Both are quantum-gravity programmes with matter-coupling-open status. Methodological precedent only — UMF does not adopt LQG or CST machinery. **Modal status:** committed (methodological precedent); informative (UMF specific quantum-gravity approach distinct) **Source provenance:** Scaffold §9.1 **First introduced:** Pre-trial **Status:** Active **Cross-references:** [CR-0023](#)

CR-JC-0011

Cluster name: GWM particle representation three-way structure **Member nodes:** A-01/Wheeler-historical, A-05/scalar-field-empirical, A-09/Hestenes-mathematical **UMF commitment supported:** GWM particle representation mainstream-precedent backing — three-way structure across historical (A-01), empirical (A-05), and mathematical (A-09) tiers **Deployment context:** Scaffold §8.1 **Annotation:** A-01 historical (Wheeler 1955 geons); A-05 empirical (boson stars LIGO/Virgo/EHT/HSC/PTA observational engagement); A-09 mathematical (Hestenes geometric algebra; Doran-Lasenby 2003 *Geometric Algebra for Physicists*). Three-way structure for GWM particle representation antecedents. **Modal status:** committed **Source provenance:** Scaffold §8.1 **First introduced:** Pre-trial **Status:** Active

CR-JC-0012

Cluster name: BEC massive-phonon mechanism source cluster **Member nodes:** Girelli-Liberati-Sindoni-2008, Marino-2019, Cominotti-2022 empirical anchor, Finazzi-Liberati-Sindoni-2012 cosmological-constant precedent **UMF commitment supported:** A-10 BEC massive-phonon framework laboratory-experimental modern realisation of wave-excitations-source-gravity Wheeler-programme commitment **Deployment context:** Scaffold §7.4 **Annotation:** Primary mechanism sources: Girelli-Liberati-Sindoni 2008 *PRD* 78:084013 (arXiv:0807.4910) gravitational dynamics in BEC; Marino 2019 *PRA* 100:063825 massive phonons. Direct empirical anchor: Cominotti et al. 2022 *PRL* 128:210401 observation of massless and massive collective excitations. Cosmological-constant-from-backreaction: Finazzi-Liberati-Sindoni 2012 *PRL* 108:071101. Mechanism nuance per v5 enrichment: finite-range gravity via massive gravitons; non-zero cosmological constant from non-condensed-atom backreaction. **Modal status:** committed (analogue-laboratory precedent); informative (specific UMF mechanism Item 6 open) **Source provenance:** Scaffold §7.4 **First introduced:** v5 (10-topic review enrichment) **Status:** Active

CR-JC-0013

Cluster name: Bullet Cluster competing-interpretations landscape **Member nodes:** Bullet-Cluster-phenomenon, Collisionless-DM-Clowe-2006, MOG-Brownstein-Moffat-2007, TeVeS-Bekenstein-2004, Refracted-Gravity-Pradyumna-Diaferio-2024, Λ CDM-collision-velocity-Mendoza-Hernandez-2014, Randers-Finslerian, Mashhoon-non-local **UMF commitment supported:** Honest-disclosure positioning of UMF gravitoelectric DM as one candidate among competing interpretations of Bullet Cluster phenomenon — no 5σ confirmation of any unique interpretation **Deployment context:** Scaffold §10.2 contested-interpretation honest-disclosure **Annotation:** Multiple interpretive frameworks contest Bullet Cluster interpretation. Bullet Cluster 8σ measurement establishes general mass-gravity-offset phenomenon regardless of framework. Honest-disclosure correction #1

(v3) captures this distinction. Synthesis-paper position: UMF joins competing-interpretations landscape as one additional candidate, none with 5σ -class confirmation. **Modal status:** committed (honest-disclosure positioning); informative (phenomenon vs interpretation distinction) **Source provenance:** Scaffold §10.2; v3 honest-disclosure correction **First introduced:** v3 **Status:** Active **Cross-references:** [CR-0028](#), [CR-0029](#), [CR-0036](#), [CD-OP-012](#)

§5 Cross-reference manifest

Outgoing cross-references from this index document, per Spec v3 §7.6:

From	Target file	Target version	Target anchor	Purpose
All bilateral edges and clusters	UMF <i>Trial</i> /S3M0S1 <i>Index1</i> ContextDefinition_v1.md	v1	various CD-* anchors	Cross-references to operational/theoretical context definitions, OP-Adj entries, OP-Item entries
CR-0042, CR-0043, CR-0044, CR-0045, CR-0046	UMF <i>Trial</i> /S3M0S1 <i>Index3</i> MetricUnits_v1.md (forward)	v1 (pending)	MU-001, MU-002	Forward-reference to Index 3 metric-units (η _B/s, M*); to be activated at Step 2c
All edges with source-provenance pointers	UMF <i>Trial</i> /S3M0S1 <i>Synthesis</i> Paper <i>Scaffold</i> v8_consolidated.md	v8	section references (§3, §4, §5, §7, §8, §9, §10, §11)	Source provenance
All edges with source-provenance pointers	VCv035 <i>Amendment10</i> adjacencyreview2026.md	(current)	§3.A-§3.K	Source provenance for AdjacencyReview2026 cycle
Edges related to v7 round	VC v0.35 Amendment 9 + Addendum	(prior anchor)	§3 sections	Source provenance for post-Jan 2026 cycle
CR-0001, CR-0035, CR-0038, CR-0039, CR-0040, CR-0050 (and related)	F-05 <i>Paper3</i> Samaddar2025 <i>AdjacencyReview2026</i> deeppass.md + Addendum	(current)	§7 honest-disclosure observations	Source provenance for Samaddar-Singh deeppass content
CR-0024, CR-0030, CR-0033, CR-0041	F-05 <i>Paper2</i> Pereira2026 <i>PostJan2026</i> deeppass.md	(current)	sections	Source provenance for Pereira deeppass content
CR-0025, CR-0026 (Whittingham edges)	F-05 <i>Paper5</i> Whittingham2026 <i>PostJan2026</i> deeppass.md	(current)	sections	Source provenance for Whittingham deeppass content
CR-0027 (Liu-Qin-Bian edge)	F-02 <i>Paper8</i> Liu2026 <i>PostJan2026</i> deeppass.md	(current)	sections	Source provenance for Liu-Qin-Bian deeppass content
All UMF-commitment edges	Introduction Letter v2	(current)	Postulate 1, Postulate 2, NOTICE	Source provenance for UMF postulates and commitments

All external corpus links per Spec v3 §7.8 (markdown filename pattern; version pinning; bulk find-and-replace at deployment).

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§6 Version history

Version	Date	Author	Notes
v1	27 May 2026 (AEST)	Claude Opus 4.7 Adaptive (Anthropic)	First populated instance per Spec v3 §11 Step 2b; 52 bilateral edges + 13 joint-citation clusters = 65 relational entries

§7 Honest-disclosure caveats at v1 close

Per modal-status preservation (CD-OP-002), the following caveats apply:

1. **Node nomenclature.** Source/Target IDs use a convention developed at population time: matrix-entry IDs (A-01, B-01, F-05, etc.); paper-within-sub-list IDs (e.g., `F-05/Samaddar-Singh-2025`); UMF-commitment IDs (`UMF-Gshock`, `UMF-A4`, `UMF-PCW`, etc.); concept IDs (e.g., `Bullet-Cluster-phenomenon`). Formal canonicalisation pending Stephen review; at v2 review naming may be tightened.
2. **Bilateral edge count (52) is non-exhaustive.** Edges populated focus on decision-relevance; trivial relationships omitted per Spec v3 §4.5 OUT scope. Particular under-coverage: A-series experimental-analogue cross-edges (A-06/A-10/A-04/A-08/A-09); F-09 Hořava-Lifshitz / GW170817 constraint; deep historical lineage edges (Bekenstein-Hawking → 't Hooft → Susskind → Jacobson → Verlinde captured as foundational only). v2 review can refine.
3. **Joint-citation cluster count target met.** 13 clusters per Scaffold §14.1 enumeration. Additional candidate clusters surfaced but not formalised: Bullet Cluster competing-interpretations landscape (now formalised as CR-JC-0013); GWM particle representation three-way (CR-JC-0011); BEC massive-phonon source cluster (CR-JC-0012). Stephen may identify additional Scaffold-deployed clusters at v2 review.
4. **Forward-references to Index 3.** Edges in §3.6 (parameter-regime) reference MU-001, MU-002 — placeholders pending Index 3 v1 population (Step 2c). Activation of these hyperlinks at Step 2c.
5. **Held-aside edges.** CR-0039 (Samaddar-Singh → Azhar-Jawad-Rani 2020) has target nodes flagged Held-aside per Amendment 10 §3.I. Edge retained for follow-up provision per Spec v3 §4.5 + Stephen 27 May AEST answer (c).
6. **Modal-status preservation re-applied.** Throughout population, "contests" reserved for actual corpus contestation (CR-0028, CR-0029, CR-0030, CR-0031); "parallel" reserved for actual independence (CR-0034, CR-0035); over-attribution flagged where present; UMF positioning under Item 5 / Item 6 / Item 7 retained as open question (modal status preserved on cluster edges).
7. **Cluster CR-JC-0013 Bullet Cluster competing-interpretations** carries multiple member nodes that overlap with bilateral edges CR-0028, CR-0029, CR-0036. The cluster is the primary representation per Spec v3 §4.8 pair-decomposition rule; bilateral edges record specific contestation relationships for finer-grained navigation.
8. **Edges where Stephen-verbatim quotes desirable but absent.** Several adjudication-triggered clusters (CR-JC-0006 Issue 2; CR-JC-0007 Issue 1) reference OP-Adj entries whose Stephen-verbatim fields are marked "[Not in available corpus]" in Index 1. Index 2 inherits this gap.

§8 Status at v1 close

First populated instance of Index 2 ready for Stephen review. 52 bilateral edges + 13 joint-citation clusters = 65 relational entries. Forward-references to Index 3 noted for Step 2c activation. Honest-disclosure caveats §7 logged.

Awaiting direction: - Approve Index 2 v1 → proceed to Step 2c (Index 3 populated) - Revise Index 2 v1 → identify changes for v2 (e.g., additional edges; refined node nomenclature; missed clusters) - Defer

End of UMFTrialS3M0S1Index2ContextRelationships_v1.md